



EXCRETORY SYSTEM — FULL MIND MAP

Biology Ch. 7.vi · Human Anatomy & Physiology · Hinglish · SSC / BPSC / BSSC / Railway

Topics in this sub-chapter:

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1. Excretion & Main Excretory Organ

- Humans mein kidneys excretory system ke main organs hain. Inka basic role blood ko filter karke unnecessary aur waste products ko body se remove karna hai. [Q10]

2. Liver — Detoxification of Alcohol

- Source ke according alcohol ka detoxification liver karta hai. Liver complex chemical reactions ke through harmful substances ko detoxify karta hai. [Q1]

3. Urine — Daily Volume, Water & Yellow Pigment

- Healthy adult human source ke according around 1.5 litres urine per 24 hours excrete karta hai; urine ka about 91–96% part water hota hai. [Q2]
- Human urine ka yellow colour urochrome pigment ki wajah se hota hai; source ise urobilin ke naam se bhi mention karta hai. [Q3]

4. Kidney Blood Flow, Filtration & Reabsorption

- Renal artery waste-containing blood ko kidneys/nephrons tak filtering ke liye le jaati hai; waste removal ke baad cleaner blood renal vein ke through kidney se bahar nikalta hai. Isliye “clean blood renal artery se wapas jaata hai” statement source mein wrong hai. [Q4]
- Bowman’s capsule se filtered liquid tiny tubules mein pass karta hai, jahan source ke mutabik much of the glucose reabsorbed hokar blood mein renal vein side return karta hai. [Q4]
- Kidneys body ke blood ke natural filters hain. Source Q8 “all the blood in the body is filtered through the kidneys” ko correct select karta hai; kidneys water-soluble wastes jaise urea/ammonium remove aur water, glucose, amino acids ka reabsorption bhi karti hain. [Q8]
- Urea blood se kidney mein separate/remove hota hai; source ultrafiltration ko renal capsule mein blood aur filtrate ke barrier par occurring process batata hai, jisse urine formation hoti hai. [Q9]
- Dono kidneys se average blood flow source ke according about 1200 cc (ml) per minute hai. Same explanation roughly 1500 litres blood purification per day aur ~1 ml urine formation per minute ka reference deti hai. [Q11]

⚠️ EXAM TRAP — Renal Artery vs Renal Vein

- ▶ Renal artery kidney ko waste-containing blood filtering ke liye laati hai; cleaner blood renal vein se kidney se nikalta hai. Q4 mein isi wajah se statement 1 wrong aur statement 2 correct hai. [Q4]

5. Dialysis & Dialyzer (Artificial Kidney)

- Dialysis blood se waste aur excess water remove karne ki process hai aur kidney failure mein lost kidney function ke artificial replacement ke roop mein use hoti hai. [Q5, Q6]
- Dialyzer ko source “artificial kidney” describe karta hai. Ye semipermeable membrane ke across flowing blood aur dialysate streams ke beech solutes aur water ka controllable transfer provide karta hai. [Q7]

6. Kidney Failure & Nitrogenous Wastes

- Kidneys fail hone par blood mein nitrogenous waste products accumulate hote hain. Source main nitrogenous wastes mein ammonia, urea aur uric acid list karta hai; urea urine ka primary component bataya gaya hai. [Q12]

7. Normal Functions of Kidney — and What Is NOT Its Function

- Source ke listed normal kidney functions mein blood-water level regulation, urea filtering aur several hormones ka secretion included hai. [Q13]
- Blood sugar level regulation ko Q13 mein normal kidney function nahi maana gaya; source explanation pancreas/insulin ko blood sugar regulation se link karti hai. [Q13]

8. Kidney Stones — Calcium Oxalate & Oxalate Crystallization

- Human kidney stones ka major/most common constituent source calcium oxalate batata hai. [Q14, Q16, Q17]
- Kidney stones urine ke chemicals se bane hard objects hain. Source Q15 stone formation ko oxalates ke crystallization se link karta hai; explanation calcium ke urine oxalate ke saath combine hone ko common calcium-oxalate stone formation batati hai. [Q15]
- Source explanation kidney stones ke four types list karti hai: calcium oxalate, uric acid, struvite aur “cystine” (source spelling). [Q15]

9. Skin as an Excretory Organ — Sweat

- Sweat skin ka excretory function bhi hai aur body temperature regulation ke liye liquid form mein produce hota hai. Source sodium, potassium, calcium, magnesium aur trace elements jaise zinc, copper, iron, chromium, nickel aur lead ko sweat se excreted batata hai. [Q18]

⚠️ EXAM TRAP — Q18 “More Than One”

- ▶ Q18 ke options mein Iron, Magnesium aur Zinc alag-alag diye hain. Source explanation in teeno ko sweat se excreted trace/electrolyte elements mein include karti hai, isliye source correct answer (E) “None of the above/More than one of the above” select karta hai. [Q18]

10. Kidney Transplantation — Immunosuppression

- Kidney transplantation mein source monoclonal antibodies ko immunosuppressor/anti-rejection role se associate karta hai. Ye immune response ke specific cells/molecules, jaise T cells ya IL-2 receptor, ko target kar sakti hain. [Q19]
- Source explanation monoclonal antibodies ko induction therapy, steroid-resistant acute rejection treatment, delayed graft function ke effects ko reduce karne aur immunosuppressive minimization se bhi link karti hai. [Q19]